

HYDROPONICS MONITORING SYSTEM

Redesign Specification

Revision 4 - Updated Electrical Architecture

Project direction: Complete and modernize the inherited prototype while keeping the existing pH and TDS nutrient-sensing approach. The redesign centers on a custom PCB, ESP32-S3 firmware, BLE commissioning, a network dashboard, and a new electrical power/control system.

Prepared for engineering design and implementation planning

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1. Purpose and Scope

This document defines the current redesign direction for the Hydroponics Monitoring System. It is based on the inherited ESP32/MQTT prototype and the design decisions agreed during the redesign review.

The project name remains a monitoring system because monitoring, configuration, diagnostics, and visualization are the primary deliverables. Local actuator control and fail-safe behavior remain part of the embedded design where they improve reliability.

Scope decision: The existing pH and TDS nutrient sensors will be retained to protect schedule and reduce integration risk. The redesign will improve their power, filtering, calibration, and PCB integration rather than replace them with new EC hardware.

2. Existing System Baseline

Inherited element	Status
ESP32 firmware	Modular C++ firmware with runtime device objects.
Dynamic device creation	Supported device classes can be created/removed from configuration without reflashing.
MQTT	Used for telemetry and actuator commands.
Backend	Universal Gateway / EdgeX-style infrastructure with MQTT and database components.
Python tools	Used for device/profile registration and commands.
Prototype state	ESP1 breadboard validated; remaining nodes and some sensor functions incomplete.
Nutrient sensing	pH 4502C and analog TDS module retained for this redesign.

3. Updated Redesign Objectives

- Design and manufacture a custom PCB around the ESP32-S3.
- Design a new electrical power/control system with external power, 2S battery backup, automatic source switchover, protected 5 V and 3.3 V rails, and filtered analog sensor power.
- Replace the DHT22 with a BME280 for air temperature, humidity, and pressure.
- Add a waterproof DS18B20 probe for reservoir water temperature.
- Retain the existing pH 4502C and TDS sensing approach and complete their calibration/interface design.
- Retain the MQ-7, HC-SR04, and ambient-light sensing functions unless later removed by scope review.
- Use explicit FreeRTOS tasks and safe inter-task communication on the ESP32-S3.
- Provide BLE commissioning and maintenance through a Python laptop application.
- Move network settings out of hard-coded firmware and into persistent configuration.
- Provide a smart network dashboard for live data, history, alarms, status, and diagnostics.
- Maintain essential local safety behavior if Wi-Fi, MQTT, or backend services fail.

4. Proposed System Architecture

Layer	Responsibility
Sensors	Measure air conditions, water temperature, pH, TDS, water level/distance, gas/CO, and ambient light.
ESP32-S3	Acquisition, filtering, calibration, device management, local logic, BLE, Wi-Fi/MQTT, and diagnostics.
Electrical subsystem	Power conversion, battery charging/protection, automatic source selection, rail filtering, and load distribution.
Actuators	Relays or MOSFET-driven pumps, valves, fans, or lights where required.
BLE desktop application	Commissioning, network setup, device management, diagnostics, and local live data.
Backend/dashboard	Remote monitoring, history, alarms, trends, node health, and safe manual controls.

5. Updated Sensor Set

Sensor	Function	Rail	Interface / note
BME280	Air temperature, humidity, pressure	3.3 V	I2C; replaces DHT22.
DS18B20 waterproof	Reservoir water temperature	3.3 V	1-Wire; use data pull-up.
Ambient light sensor	Light level	3.3 V	Analog; local decoupling and optional signal filtering.
MQ-7	CO/gas sensing	5 V	Heater load dominates sensor-side 5 V current.
HC-SR04	Reservoir distance/level	5 V	Digital; ECHO must be level-shifted/divided for ESP32.
pH 4502C	Nutrient-solution pH	5 V analog branch	Sensitive analog measurement; filtered supply and calibrated input.
TDS module	Dissolved-solids estimate	5 V analog branch	Sensitive analog measurement; filtered supply and calibrated input.

Sensor replacement: BME280 replaces the DHT22. DS18B20 is added specifically for water temperature. The pH and TDS hardware remain in scope to avoid a larger sensor redesign.

6. New Electrical Power and Control System

The electrical subsystem is a major redesign objective. The final PCB will implement the agreed power architecture directly; this specification describes its behavior rather than reproducing the electrical diagram.

6.1 External Supply and Battery Charging

The system uses an external USB-C power source. The selected 2S charging module accepts 5 V input and internally boosts the voltage to the 8.4 V required to charge a 2S lithium battery. The module is rated for high input current; the actual charging current will depend on the USB-C source and cable capability.

- External source: 65 W USB-C charger, used at the 5 V input required by the charging module.
- 2S charger: 5 V USB-C input, internal boost to 8.4 V charge voltage.
- Battery: 2S LiPo, 7.4 V nominal, 8.4 V fully charged, approximately 2600 mAh.
- Do not rely on the charger module as the battery protection device.

6.2 Battery Protection

The battery pack is protected by a separate 2S BMS/protection board. The board provides over-charge, over-discharge, over-current, and short-circuit protection. The charger and system load must be connected through the protected pack terminals according to the BMS wiring.

- Current design target: 2S, approximately 5 A protection board.
- Cell balancing must not be assumed unless the exact board documentation confirms it.

6.3 Automatic External/Battery Switchover

The battery path is regulated to 5 V and switched with an IRF4905 P-channel MOSFET. External power is used as the normal source. When external 5 V is present, the MOSFET gate is held near the source potential and the battery path is off. When external power disappears, a gate pull-down resistor forces the gate low, making VGS negative and turning the battery path on.

- IRF4905 gate pull-down: approximately 100 kOhm to ground.
- A small series gate resistor may be used if required for transient control.
- The IRF4905 is acceptable for the prototype power-path design, but its on-resistance at approximately -5 V gate drive must be verified experimentally because the strongest datasheet RDS(on) figure is specified at a larger gate magnitude.
- 1N5822 Schottky diodes provide source isolation/OR-ing; their forward drop must be included in the 5 V rail budget.
- MOSFET source/drain orientation must be checked against the intrinsic body diode before PCB release.

6.4 Main 5 V Node

The selected source feeds the main 5 V node. This node supplies the 5 V peripherals and the input of the 3.3 V regulator. Bulk and high-frequency decoupling are provided near the main node to absorb load steps and switching noise.

Function / load	Recommended treatment
Main 5 V node	Bulk capacitor around 470 uF plus local 10 uF and 100 nF decoupling as starting values.
MQ-7	Dedicated 5 V branch with local 100 nF, 10 uF and additional bulk capacitance as required by heater transients.
HC-SR04	5 V branch with local 100 nF and approximately 10 uF decoupling.
pH + TDS	Do not share the noisy branch directly; feed from a ferrite-filtered 5 V analog rail.

The capacitor values above are implementation starting points. Final regulator stability and capacitor type/value must comply with the selected regulator datasheets.

6.5 Filtered 5 V Analog Rail

The pH and TDS modules are powered from a quieter analog 5 V branch derived from the main 5 V node through a ferrite bead. The purpose is to reduce high-frequency noise from the ESP32, switching regulators, MQ-7 heater, and other digital/high-current loads.

- Ferrite bead: select a two-terminal PCB ferrite bead with low DC resistance and adequate current rating.
- A value in the general range of 120-600 Ohm impedance at 100 MHz is a reasonable starting class; final choice is verified from actual noise measurements.
- Use 100 nF and approximately 10 uF decoupling on the filtered side, placed physically close to the pH/TDS power branch.
- The ferrite bead is not a DC series resistor; its useful impedance is mainly at high frequency.
- Keep MQ-7 heater current and other noisy loads off this filtered analog branch.

6.6 3.3 V Rail

A dedicated 3.3 V regulator supplies the ESP32-S3 and 3.3 V sensors. The regulator must tolerate ESP32 Wi-Fi/BLE current bursts without brownout.

3.3 V load	Local decoupling / note
ESP32-S3	100 nF plus approximately 10 uF close to the module; additional bulk on the rail as needed.
BME280	100 nF close to VCC; 1 uF local capacitor may be added.
DS18B20	100 nF close to the connector/device where practical; 1-Wire data pull-up is separate from rail filtering.
Ambient light sensor	100 nF close to the device; 1 uF local decoupling may be added.

- No series resistor is placed in the ESP32-S3 power rail.

- The 3.3 V regulator should be sized with comfortable transient margin; approximately 1 A capability is preferred for the controller rail.

6.7 Grounding and Placement

- Use a solid ground reference and avoid forcing high-current heater or actuator return current through sensitive analog return paths.
- Place decoupling capacitors close to the devices they support.
- Keep the pH/TDS analog area away from switching regulators, the MQ-7 heater path, and high-current actuator traces.
- Keep the ferrite bead and its post-filter capacitors physically close to the analog sensor power branch.
- Provide test points for main 5 V, filtered 5 V analog, 3.3 V, battery voltage, and ground.

7. Custom PCB Requirements

- ESP32-S3 module implementation with RF keep-out respected.
- USB-C/external power interface consistent with the selected charger architecture.
- 2S charger module interface and 2S BMS connection.
- Battery connector with clear polarity and protected routing.
- Automatic power switchover stage using the IRF4905 and Schottky isolation path.
- Main 5 V regulator / distribution and 3.3 V regulator.
- Ferrite-filtered 5 V analog branch for pH and TDS.
- Sensor connectors for BME280, DS18B20, ambient light, MQ-7, HC-SR04, pH, and TDS.
- Required logic-level protection/dividers for 5 V sensor outputs into ESP32 pins.
- Relay/MOSFET actuator outputs where required by the final system.
- Programming/debug access, reset/provisioning buttons, status LEDs, and test points.
- Adequate trace width, connector rating, thermal design, and protection for expected current.

8. ESP32-S3 Firmware Architecture

The redesign keeps the useful runtime device-management concept but organizes the application explicitly around FreeRTOS tasks.

Task / subsystem	Responsibility
Sensor acquisition	Timed sensor reads, validity checks, filtering, calibration.
Local monitoring / safety	Thresholds, alarms, safe states, watchdog-related logic.
Actuator manager	Validated output commands and safe boot/reset behavior.
Wi-Fi / MQTT	Connectivity, telemetry, remote commands, reconnect.
BLE	GATT server, secure pairing, provisioning, local maintenance.
Device manager	Create/update/delete supported device instances safely.
Diagnostics	Fault logging, system state, health information.

Queues, mutexes/semaphores, event groups, and task notifications should be used where appropriate. Core affinity should only be applied when it solves a measured timing or interference problem.

9. BLE Commissioning and Python Desktop Application

The Python application renders the interface locally on the laptop. The ESP32-S3 exposes configuration, telemetry, status, and device-management operations over BLE GATT.

- Discover and connect to nearby hydroponics controllers.
- Pair and bond using encrypted BLE communication.
- Configure Wi-Fi credentials, MQTT/backend address, and node identity.
- Display live diagnostic and sensor data.
- View, add, update, and remove supported runtime device instances.
- Display validation errors and operation acknowledgements.
- Provide a commissioning mode that is deliberately enabled rather than permanently open.

10. GATT and Dynamic Device Management

Service	Typical operations
Telemetry	Notify sensor/status packets.
Network configuration	Read/write SSID, MQTT address, node name; password write-only.
Device management	Read device list; add/update/delete requests; result notifications.
Diagnostics/control	Alarm/status notifications and authorized actuator commands.

Device-management requests should be passed from the BLE callback to the Device Manager through an RTOS queue. The BLE stack should not directly delete or reconfigure an object being used by another task.

- Use stable internal device-type IDs rather than depending only on profile-name strings.
- Validate GPIO capability, conflicts, required pin count, duplicates, and configuration ranges.
- Creating another instance of an already-supported sensor should not require reflashing.
- Adding a completely new sensor class still requires firmware development and recompilation.

11. Network Dashboard and Backend

BLE is primarily for local commissioning and maintenance. Normal remote monitoring uses Wi-Fi/MQTT and the backend.

Dashboard area	Content
Live measurements	BME280 values, DS18B20 water temperature, pH, TDS, level, light, MQ-7.
History	Time-series trends for the important environmental and water measurements.
System health	Node connectivity, MQTT status, battery/power status where measured, faults.
Alarms	Sensor invalid, low water, communication loss, actuator timeout, power events.
Calibration	pH/TDS calibration state and service information.
Manual functions	Only safe, authorized controls and configuration.

12. Local Fail-Safe Behavior

- Continue essential sensor acquisition during network loss.
- Keep configured safety rules active without MQTT.
- Apply safe actuator states at boot/reset.
- Prevent excessive pump-on time where actuators are controlled locally.
- Detect and flag impossible or missing sensor readings.
- Record/report power-source transitions and brownout events where practical.
- Resume telemetry cleanly after Wi-Fi/MQTT recovery.

13. Electrical and System Validation Plan

Area	Minimum validation
USB-C charger path	Verify 5 V input behavior, real input current, charge current, thermal rise, and cable/source limits.
2S battery/BMS	Verify correct protected terminals, over-current behavior, low-voltage cutoff, and pack polarity.
Power switchover	Measure external-to-battery and battery-to-external transfer; confirm no reset/brownout of ESP32-S3.
IRF4905	Measure VGS, voltage drop, and temperature at expected battery-path current.
1N5822 path	Measure forward drop at operating current and verify resulting main-rail voltage.
5 V main rail	Check ripple and transient response with MQ-7 and other loads switching.
5 V analog rail	Compare pH/TDS noise before and after ferrite filtering.
3.3 V rail	Verify ESP32 Wi-Fi/BLE burst stability and regulator temperature.
BME280 / DS18B20	Verify communication, accuracy, and long-run stability.
pH / TDS	Calibrate with reference solutions and quantify repeatability/drift.
BLE	Verify secure pairing, reconnect, unauthorized-write rejection, and device management.
Network	Verify Wi-Fi credential changes and MQTT reconnect behavior.

14. Open Engineering Decisions

- Confirm the exact charger-controller IC on the 5 V-to-8.4 V 2S charging module from board markings.
- Confirm the exact BMS model and whether it provides cell balancing or protection only.
- Select the final 5 V and 3.3 V regulator parts and verify their stability capacitor requirements.
- Select the final ferrite bead after expected current and measured noise are known.
- Confirm the exact inherited TDS module and its calibration transfer function.
- Confirm final actuator loads and decide relay versus MOSFET channels.
- Determine whether the existing backend is retained unchanged or simplified.
- Finalize BLE packet format, UUID map, and persistent configuration schema.

15. Deliverables

1. System requirements specification.
2. Updated electrical design and component selection.
3. Custom PCB schematic and layout.
4. Power-budget and battery-runtime calculations.

5. ESP32-S3 firmware architecture and FreeRTOS task design.
6. BLE/GATT protocol specification.
7. Python commissioning/maintenance application.
8. Dashboard/backend specification and implementation.
9. pH/TDS calibration procedure.
10. Electrical, firmware, BLE, and system validation report.

16. Status Note

This revision records the current agreed redesign direction. Values such as capacitor sizes, ferrite impedance, regulator ratings, and charger/BMS model identifiers are design starting points or current hardware assumptions and must be verified against the exact parts used before PCB release.

Key revision changes: BME280 replaces DHT22; DS18B20 is added for water temperature; pH and TDS are retained; the new battery-backed electrical power/control system is now a primary project objective; the electrical architecture is described in text and tables rather than redrawn.